University of Florida, Department of Mathematics

Analysis PhD Exam, September 2002

I

State the following theorems.

1. The Egorov theorem.
2. The Lebesgue theorem.
3. The dual of $L^p(\mu)$, $1 \le p < \infty$.
4. The Fubini theorem for $\mu \times \nu$-integrable functions.
5. The Riesz representation theorem (of linear continuous functionals on the space $C(K)$ of continuous functions on a compact space K).

II

State and prove one of the following 2 theorems.

1. The representation of the dual of $L^1(\mu)$.
2. The Fubini theorem for $\mathcal{S} \times \mathcal{T}$-measurable functions.

III

Solve the following problems.

Let (X, Σ, μ) be a measure space and E a Banach space.

1. Assume μ is finite. Let $\mathcal{F} \subset \Sigma$ be a sub-σ-algebra. Prove the existence of the conditional expectation $E(f \mid \mathcal{F})$ for functions $f \in L^1_E(\mu)$.
2. Let $\mathcal{R} \subset \Sigma$ be an algebra generating Σ and $f \in L^1_E(\mu)$. Prove that there is a sequence (f_n) of $\mathcal{R}$-step functions converging to f, μ-a.e., and in the mean.
3. Let $\mathcal{R} \subset \Sigma$ be an algebra generating Σ and $f \in L^1_E(\mu)$. Prove that if $\int_A f\, d\mu = 0$ for every $A \in \mathcal{R}$, then $f = 0$, μ-a.e.
4. Assume μ is finite and let $f_n \in L^1_E(\mu)$, $n = 1, 2, \ldots.$ Prove that if $f_n \to f$ uniformly, then $f \in L^1_E(\mu)$ and $f_n \to f$ in the mean.
5. Let $(X, \mathcal{S})$, $(Y, \mathcal{T})$ be measurable spaces, and let $\mu : \mathcal{S} \to \mathbb{R}$ and $\nu : \mathcal{T} \to \mathbb{R}$ be σ-additive measures. Prove the existence of the product measure $\mu \times \nu$ and the equality $|\mu \times \nu| = |\mu| \times |\nu|$.

Note. Write complete computations and explanations. Do not use arrows or other unnecessary signs. Use words for explanations. Write complete sentences. Write complete statements of theorems (including the framework).